Management of change, innovation and growth

M a r c o M a g n a n i

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Survival instinct *(fear factor) helps to* ***manage change*** *and stimulates* ***innovation***

Managing change *and being* ***innovative*** *may be a matter of* ***survival***

Change:

threat or opportunity?

We are currently facing disruptive changes in many domains

- Geopolitics
- Environment
- Energy
- Economics and Society
- Technology

We are facing radical changes

Geopolitics

- Wars and political instability
- Change of geopolitical balance
  - "New World Order"
- Risk of new "cold war"
- Risk of decoupling
- Multipolar world
- Rise of regional powers
- International terrorism
- Migration flows
- Polycrisis and permacrisis

We are facing radical changes

<u>Environment</u>

- Climate change
- Global warming

<u>Energy</u>

- Green transition
- Renewable sources
- Energy storage
- Next generation nuclear power
- Shale gas (fracking)
- Drilling in the Artic

We are facing radical changes

<u>Economy and society</u>

- Globalization vs. protectionism
- Decoupling
- Risks of economic crisis
    - inflation, recession, secular stagnation
- Demographic trends
- Diversity and multicultural society
- Middle class struggle
- Increasing inequality
- Middle income trap

We are facing radical changes

Technology

- Robots and automation
- UAV-Unmanned autonomous vehicles (air, land, sea; people and goods)
- 3D printing and additive manufacturing
- Digitalization
- e-Commerce
- Internet of Things (IoT)
- Big Data
- Cloud computing and storage
- AI and machine learning
- Augmented (AR) and virtual reality (VR)

We are facing radical changes

Technology *(continued)*
- Blockchain
- Quantum computing
- Nanotechnologies e nano-materials
- Biotechnologies

Others
- Genoma engineering
- Micro and nano sensors
- Neurobionics
- Space economy

Impact of new disruptive technological change and innovations…

- …on **economic growth**
  - Sustainability constrains
    - ✓ environmental
    - ✓ demographic
    - ✓ food
    - ✓ energetic
    - ✓ social and institutional
- …on **labour**
  - Relations human - machine
  - Future of work

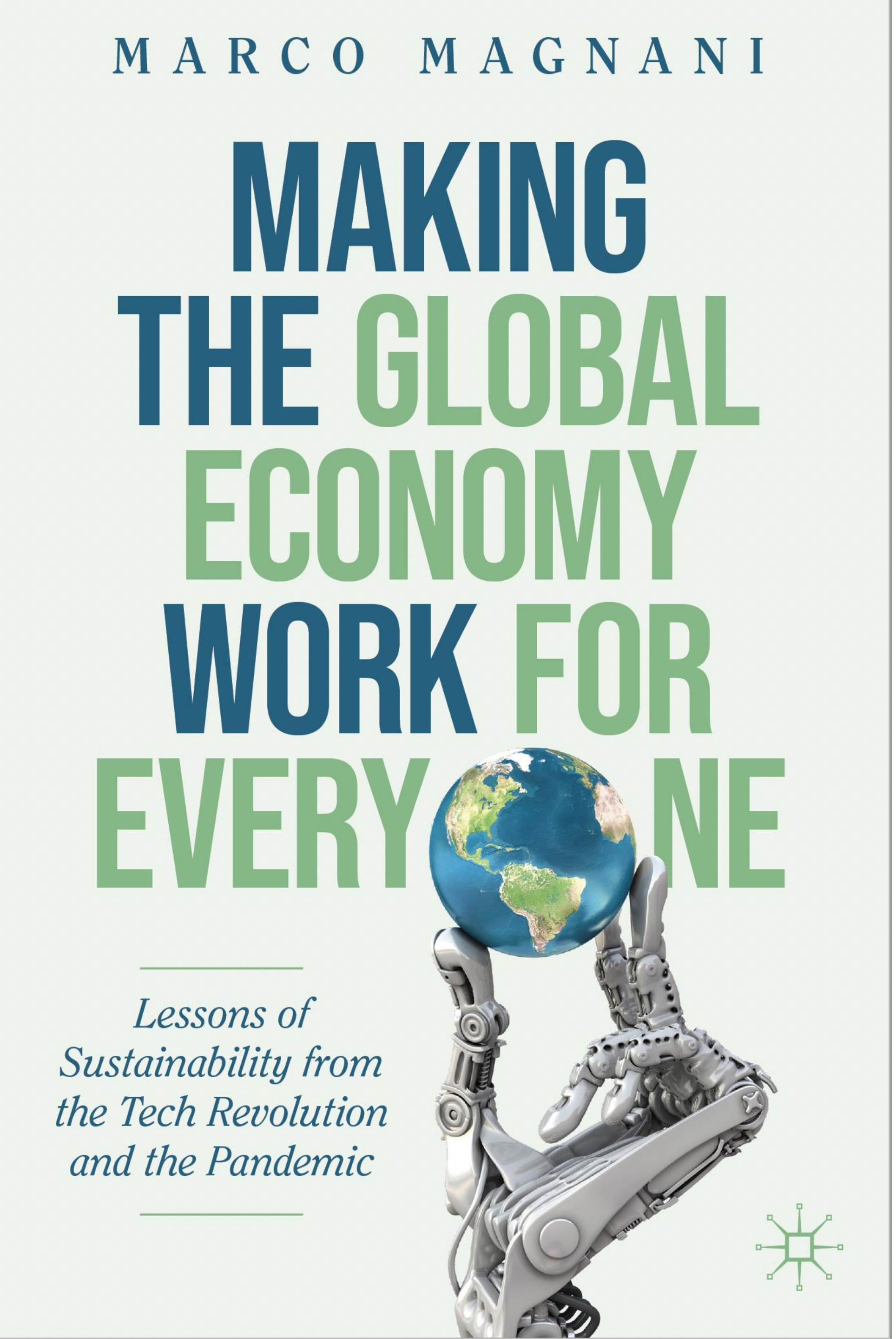
Fatti non foste a viver come robot

Crescita, lavoro, sostenibilità: sopravvivere alla rivoluzione tecnologica

di Marco Magnani

UTET

(272 pagine, €15)



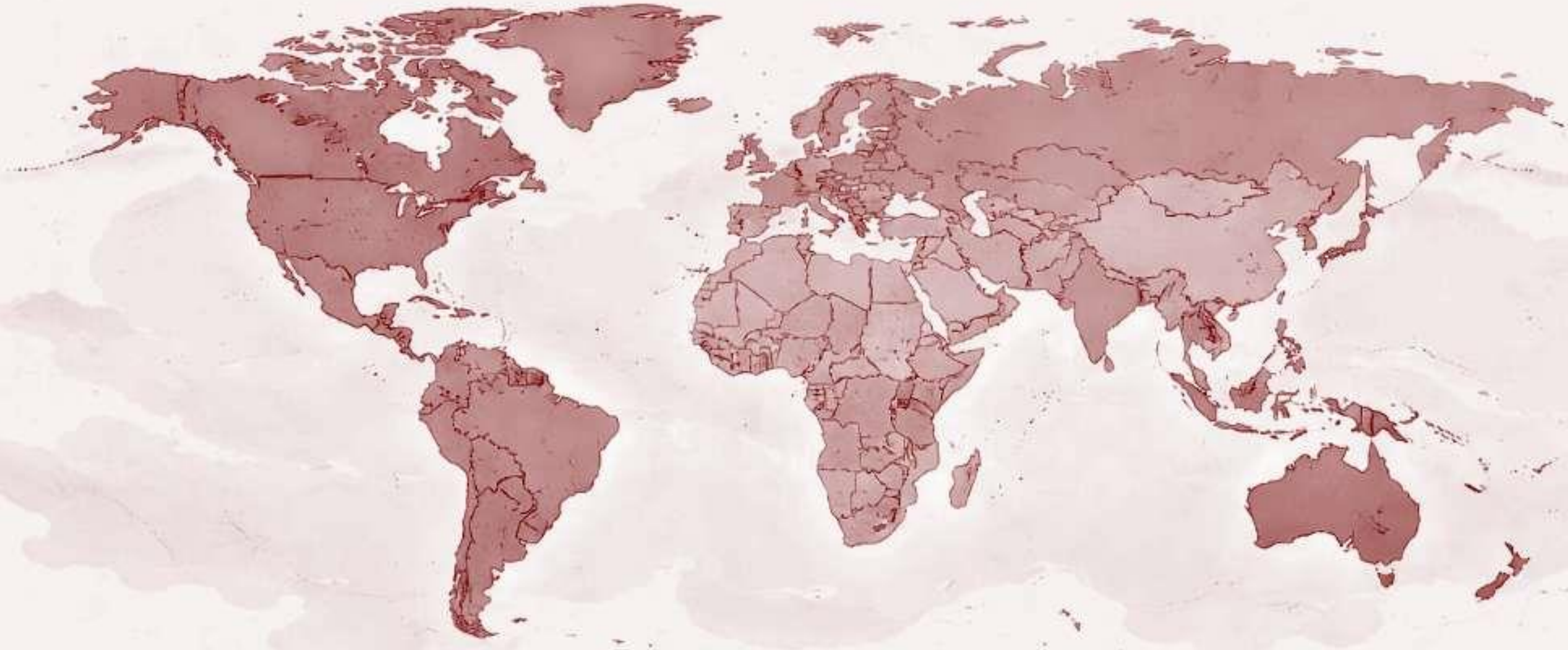
Making the global economy work for everyone

Lessons of sustainability from the tech revolution and the pandemic

(Palgrave Macmillan, 2022)

Change of the world balance: a “new global order”?

Major Global Conflicts to Watch in 2026



Increasing number of conflicts

- Highest number of conflicts since 1945
  - **120-130 active conflicts** worldwide
  - approximately **60 countries** involved in warfare
  - **over 120 non-state armed groups** engaged in fighting
  - significant increase over the past 15 years
  - **1 out 6** lives in areas with active conflicts

Source: Conflict Index 2024. The Armed Conflict Location & Event Data Project (ACLED).

Third world war «in pieces» ?

- Russia – Ukraine
- Israel – Hamas, Hezbollah, Houti
- Israel / Usa – Iran
- Libya
- Syria
- Kosovo
- **Yemen** ✔
  - Gulf di Aden
  - access to Red Sea
- **Sahel region** ✔
  - 2020-23: 8 coup d'état in Guinea Conakry, Ciad, Niger, Gabon, Mali (2), Burkina Faso (2)
- **Azerbaijan – Armenia** ✔
  - Nagorno-Karabakh
- Pakistan - Afghanistan
- Iran – Pakistan
- Senegal
- Ethiopia
  - Tigray
  - Tensions with Egypt
- Somalia
- Sudan & South Sudan
- Mozambique
- Dem. Rep. of Congo
  - M23, Ruanda
- Haiti
- Mexico
- Ecuador
- Venezuela
- Myanmar
- Taiwan

The spirit of commerce
is incompatible with war

Immanuel Kant
Perpetual Peace, 1795

No two countries that both have a McDonald's have ever fought a war against each other

Thomas Friedman
The New York Times, 1996

- Peace and stability are key pre-requisite for trade, investment, innovation and economic growth

- **Not always trade and investment can guarantee peace and stability**

The great disconnect

The many dimensions of the «**great disconnect**»

- Trade and investments
- Currency and payment systems
- Technology (chips, “splinternet”, etc.)
- Strategic raw materials (rare earth)
- Energy
- Culture
- Science and medicine
- Geopolitics
- International cooperation

1. Weak **international institutions**
   - ✓ Crisis of international cooperation
2. **Autocracies** and **illiberal democracies** with the goal of disrupting current world balance
   - **“The quartet of chaos”**
   - The new positioning of the **U.S.A.**
3. Crisis of globalization
   - **Slowbalization**

The big promises of globalization:
the 3 Ps

1. **Prosperity**

2. **Progress**

3. **Peace**

Is there a relationship between globalization and democracy?

Globalization
unstoppable and **irreversibile**

The enormous benefits of **modern globalization** have spread the (illusory) belief that it could guarantee

- continuous economic growth
- easy circulation of innovation, technology and knowledge
- an end to all opposition and conflicts

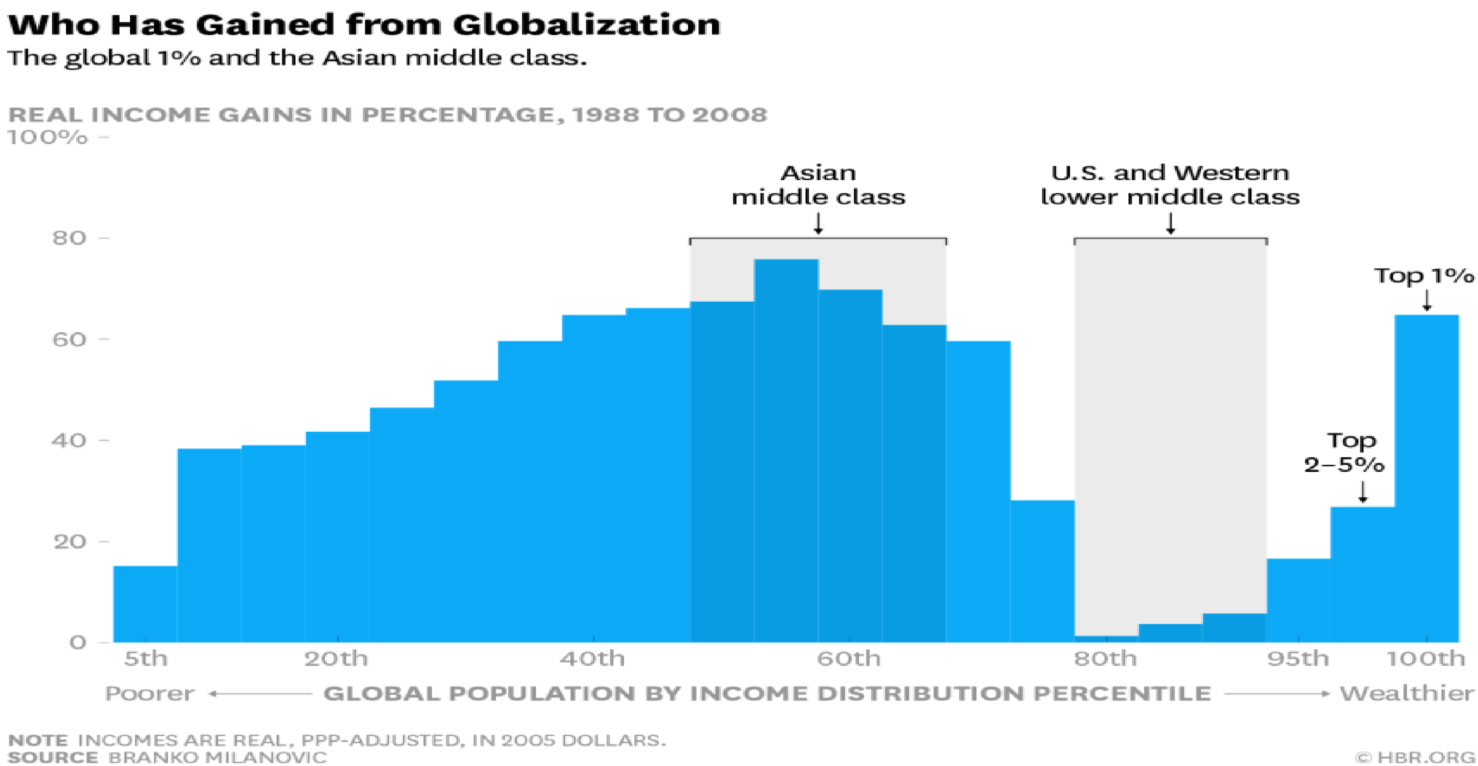
The end of history?

Distribution distortion

- **Distortion in the distribution of income**
  - **Within countries**
    - Increasing inequality?
  - **Among countries**
    - Convergence or divergence?

Distribution distortion within countries: winners and losers

- Asian middle class and top 1% are the winners
- Western middle class has suffered the most



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Convergence or divergence: the gap between rich and poor

Has the world income gap narrowed over time?

- **Convergence** between income levels of high-income countries and some previously middle-income and low-income countries (fast-growing countries)
- Very low growth rate of some of the poorest countries and resulted in a wider income gap (**divergence**) between high-income and slow-growth countries

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Limitation of national sovereignty

- Great degree of **interdependency** among countries increase exposure to external shocks
  - Increases impact of spill-overs
  - Increases countries vulnerability
- This limits **national sovereignty**
  - Ability to issue laws and standards that reflect the national economic and social goals

Impact on non economic variables

Culture
Local identity
Environment

<u>Risks (real or perceived)</u>
McDonaldization
Loss of local identity
Negative impact on environment

Slowbalization

- **Slowdown**, not turnaround
- **Reorientation** of economic relations

The level of economic integration has not decreased in absolute terms

Slowdown and reorientation are opportunities but also a sign globalization fatigue

Have the engines of globalization jammed?

- Innovation (transportation and tlc)
- General climate of peace and relative stability
- Political-institutional framework conducive to international cooperation
- U.S.-China relationship
- Impact (real or perceived) on non-economic variables
  - Culture, local identity, environment
- Excesses, limits, distortions
  - Distortions (real or perceived) in distribution
  - Limits on national sovereignty

Growing anti-globalization sentiment

De-globalization drivers

1. Economic rationale
2. Exposure to external shocks
3. Political (domestic) rationale
4. Geopolitical rationale
5. Conflict and uncertainty

De-globalization drivers:
1. economic rationale

- Increasing cost of labour in emerging economies
- Increasing automation
- Change in consumer preference
- Other issues with delocalization:
  - Quality of production
  - Suppliers' ethics

- Reshoring
- Near-shoring

De-globalization drivers: 2. exposure to external shocks

- In a world with strong economic interconnections and very long global supply chains, an external shock hitting only one of the links easily becomes systemic

The black swan theory

*An **unexpected** and **unpredictable** event of **large magnitude** that has **systemic and global impact** and whose consequences have a dominant role in history*

(Nassim Nicholas Taleb)

Examples of black swans in history

- The Spanish Conquistadors and the Aztecs
- The sinking of the Titanic
- The Fukushima accident
- 9/11 World Trade Center terrorist attack
- The collapse of the Soviet Union

Impact of Covid-19 on global supply chain

- Affected sectors relying on Wuhan (**Hubei** province)
- **Automotive**
  - ✓ Feb. 4: Hyundai (5th world carmaker: shuts down of all factories in South Korea for lack of parts from China
  - ✓ Nissan, Toyota, Honda, Ford: extended shutdown of joint-venture operations in China
  - ✓ Feb 7: FCA stops production in European plant due to issues with auto parts supply
- **Technology**
  - ✓ Apple, Xiaomi, BOE: issues with factories in Wuhan
  - ✓ Foxconn loses 10% on the stock market in a few days

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Global supply chains: the iPad3

- **Apple works with suppliers in 43 countries and 6 continents**
- Apple supply chain gets even more complicated when broken it down to raw materials

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Advantages of global supply chains

- **Cost**
  - lower labor cost
  - lower production and operating costs
- **Flexibility**
  - increases flexibility
  - competition among suppliers
- **Innovation** potential

Maximization of efficiency

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Disadvantages of global supply chains

- **Quality risk**
- **Loss of control**: difficulty to manage communications and oversee technical aspects of production process
- **Longer lead times**: production time can be quick but lead time can be long due to shipping (this requires forward planning)
- **Reputational risk**: risk of exposure to unethical conduct (e.g., modern slavery), risk of damaging brand
- **Strategic risk**: risks of political instability and social unrest in suppliers' countries ✔
- **Fluctuations in exchange rates**

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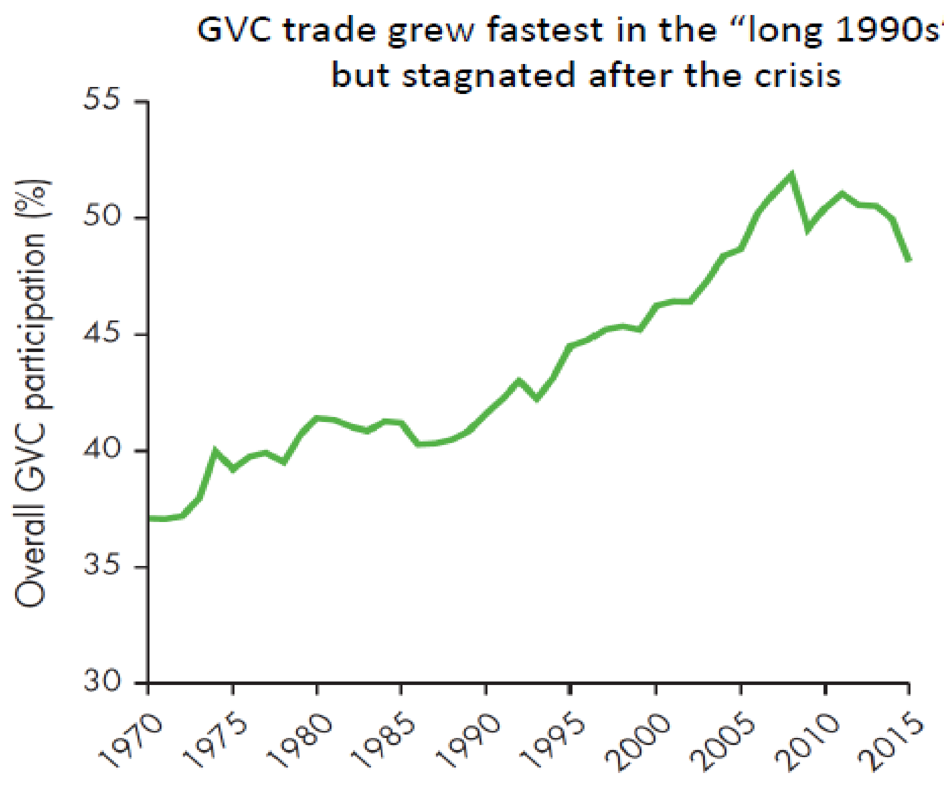
The slowdown of Global Value Chains trade

The Slowdown of GVC Trade



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Efficiency vs. Resilience

The new supply chains

- **The new trend will be to place more attention to the localization of supply chains**
    - Less vulnerable
    - Shorter
    - Closer to end-markets

De-globalization drivers: 3. political (domestic) rationale

- Rise of nationalism
  - Made in China 2025 (Xi Jinping)
  - Make in India
  - America First (Trump)
  - Build America Back and IRA (Biden)
- Rise of protectionism and increasing trade policies
  - Trade tariffs and other other barriers
  - Trade wars

De-globalization drivers: 4. geopolitical rationale

- Using for geopolitical goals:
  - energy
  - strategic raw materials and rare earths
  - agricultural commodities, water
  - technology
  - trade and trade surplus
  - currency and payment networks
  - investment (infrastructure, foreign debt)
  - migration flows
  - medical products, vaccines

De-globalization drivers: 5. conflict and uncertainty

- Uncertainty kills trade, investment, innovation
- Risk vs. Uncertainty

What is the pre-condition for international trade and investment to prosper, for international growth and for innovation?

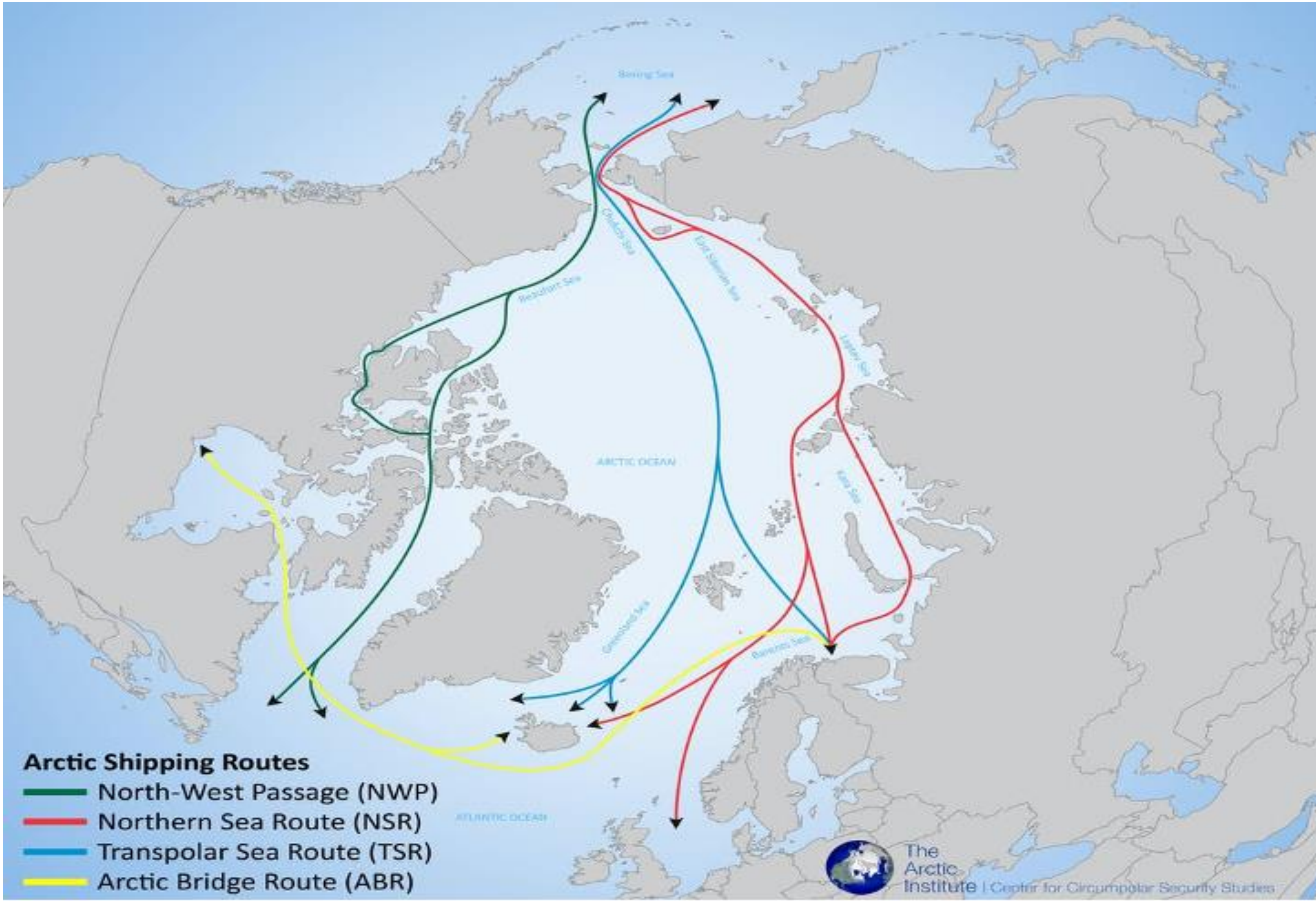
Security of trading routes and political stability have always been key prerequisites for international trade and investments (and for economic growth and innovation)

Are routes of international trade and investment secure today?

What is the current level of political stability worldwide?

What are some of the new routes of trade?

The new Arctic routes



The Globalization Paradox

1. Economic **globalization**
2. Ability of Governments to make decisions (**national sovereignty**)
3. Compliance of collective decisions with preferences of the majority of the people (**democracy**)

are not simultaneously pursuable

The new trilemma

1. **Combat climate change**

2. **Reduce global poverty**

3. **Boost the middle class in advanced economies**

“Under current policy trajectories, any combination of two goals appears to come at the expense of the third”

Trends and risks
in the new fragmented world

The **trends** of a fragmented world

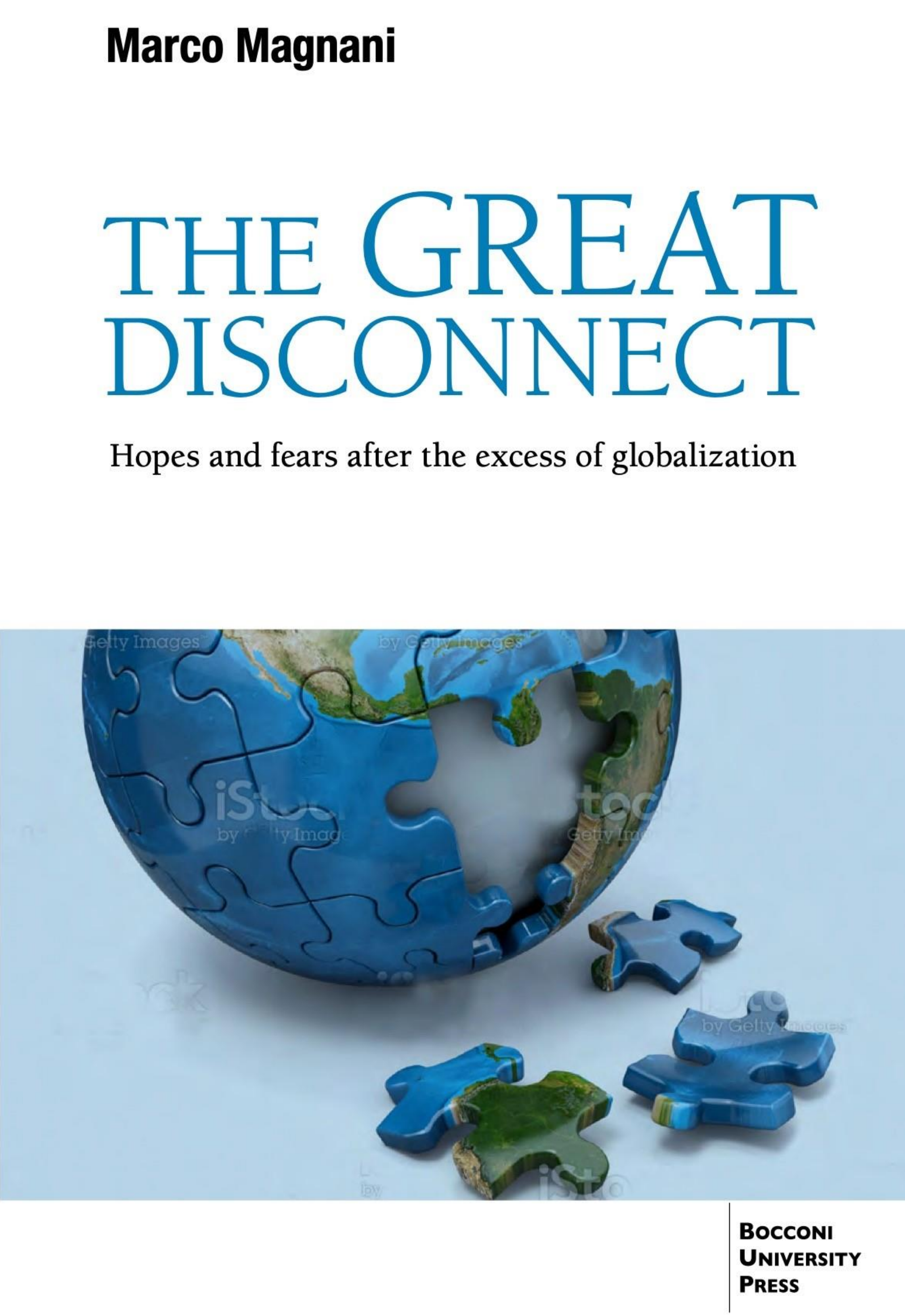
1. Strong regionalization
   - Coexistence of regionalization and globalization increase complexity of international scenario
2. Increasing prevalence of politics over economics
   - National security and geopolitical interest drive economic decisions
3. Increasing volatility in international relations
   - Free agents

Economic impact
of the fragmented world

- Economic cost (less efficiency)
- Inflationary trend
- Lower innovation
- Increased role of States
- More debt

The **risks** of the fragmented world

- Fragmentation
  - increases the **complexity** of international relations
  - risk of "**polycrisis**" (and "**permacrisis**")
  - has often brought **geopolitical chaos** and conflict
- Historically, periods of openness and strong integration have been followed by phases of fragmentation with closure and loosening of international ties



The Great Disconnect

Hopes and fears after the excess of globalization

Bocconi University Press

Il Grande Scollamento

Timori e speranze dopo gli eccessi della globalizzazione

Bocconi University Press

How to manage continous disruptive change?

① Is change bigger today than in the past?

② What makes change a threat or an opportunity?

③ How can change be successfully managed?

Managing change is more difficult today than in the past

- Disruptive change is more frequent
- We are influenced also by change happening far away
- Receiving real-time information about all changes occurring everywhere, makes it more difficult to rank priorities and make decisions

Initial positioning is important

- Discovery of America (12 October 1492)
  - Spain, Portugal, England, Atlantic powers
  - Venice and other Mediterranean powers
- Fracking (hydraulic fracturing and horizontal drilling)
  - Shale gas extraction
- Global warming
  - Impact on agriculture and fishing
  - New trade routes and natural resources

Initial positioning
may mean a lot of different things

- Geographic location
- Natural resources availability
- Market share, financial position
- Family / social and economic status
- Professional situation
- Age

Being at the right place helps…

…however **initial positioning** is not always sufficient

Toys R Us

Perception of change depends also on its **management**

Fukushima, 11 March 2011

Perception of change depends also on its **management**

- Angela Merkel (June 2011) announces suspension of domestic nuclear activity and gradual decommissioning of power plants with negative impact on energy prices for German industry
  - **Change seen as a threat to be defended against**
- Italian Institute of Technology (IIT) in Genoa seizes the momentum to invest further in the development of **quadruped robots**, that replace humans in dangerous situations more efficiently than biped robots
  - **Change becomes an opportunity**

Managing change

- In a rapidly changing world the most important skill is the ability to manage change
  - country, region, enterprise, individual
- Managing change can be a compelling competitive advantage

It is not the strongest of the species nor the most intelligent that survives.

It is the most adaptable to change

Charles Darwin

The only sustainable competitive advantage is the ability to learn and to change more quickly than others

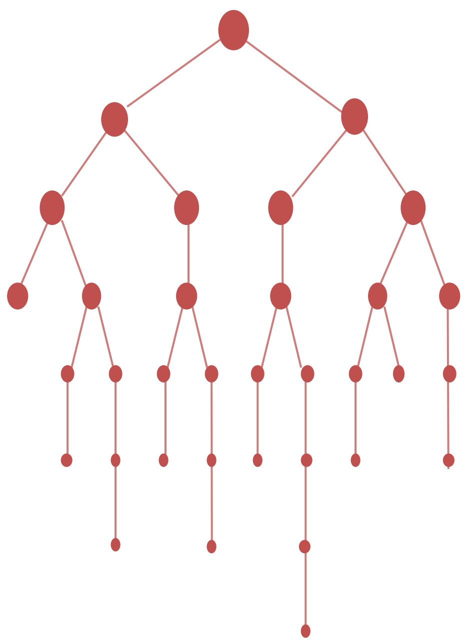
Phil Kotler

Learn

Adapt

COMMAND and CONTROL

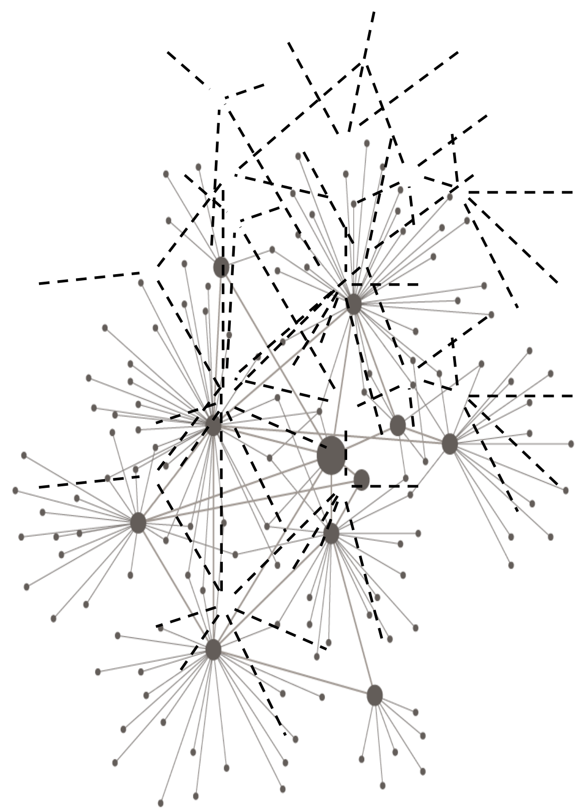
(generation X)



Static Hierarchies

LEARN and ADAPT

(generation Y & Z)



Dynamic Networks

Thinking framework and organizational framework

Different (thinking and organizational) models

Static hierarchies

- Command and control
- Information scarce
- Compete to win
- Individual productivity
- Focus on planning ahead
- Efficiency of process

Dynamic networks

- Learn and adapt
- Information abundant
- Collaborate to win
- Collective value creation
- Experiment, learn and respond
- Effectiveness of outcomes

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How to manage change?

How to manage change?

- **Suffering**
- Bend over, hoping to survive but taking the risk of being overwhelmed
- Typically when change is greatly **underestimated** or **unexpected**

✓ *Kodak, ToysRus, Detroit, great encyclopedia*

How to manage change?

- **Resistance**
- Seek protection from risk
  - insurance
  - hedging
  - defensive diversification
  - financial soundness/reserves

✓ *Blockbuster, Blackberry, Sony, Nokia (cell to smartphone), taxis vs. Uber*

Never underestimate the competition!

Microsoft CEO Steve Ballmer reaction to the iPhone

2007
https://m.youtube.com/watch?v=nXq9NTjEdTo

https://www.cultofmac.com/501138/apple-history-steve-ballmer-iphone-freakout/?utm_content=cmp-true (0-1.30)

2014
https://charlierose.com/videos/28129 (19.30-23.00)

https://www.youtube.com/watch?v=v9d3wp2sGPI

How to manage change?

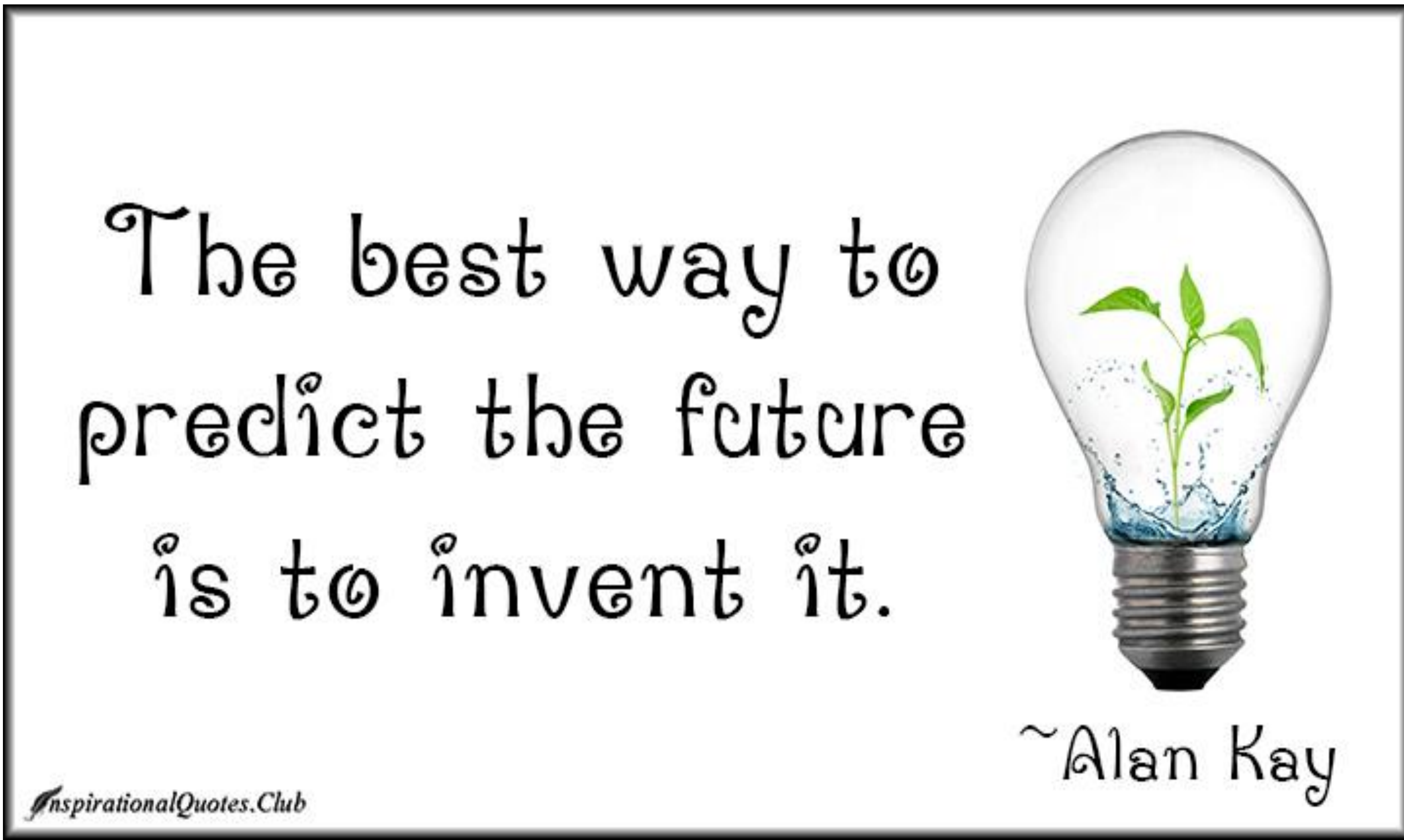
- **Resilience**
- Ability to recover from or adjust easily to misfortune or change
- Anticipate / predict change
- Ride change
- **Making change an opportunity**
  - ✓ *IBM Global Services, Nokia (paper mill, rubber, cables to communication, cell phones), Fujifilm, Barilla (Mulino Bianco), Ferrero, Pittsburgh, Torino, Dutch flower district, Rotterdam vs. water*

How to manage change?

- **Promoting** change
- Be proactive actors of change

  - ✓ Apple (smartphone-touch, pc-design)
  - ✓ CNN (all news channel)
  - ✓ Ford (Model T, assembly line)
  - ✓ Singapore (economic positioning)

Be proactive actors of change



Be proactive actors of change

There is a world where people don’t just let things happen.

They make them happen

Sergio Marchionne

How to manage change?

Suffering

Resistance

Resilience

Promoting

Managing change is primarily a matter of mindset

How to ride the wave of change without being overwhelmed by it?

L'Onda Perfetta

Cavalcare il cambiamento senza esserne travolti

Edizione

IlSole24Ore

In edicola e in libreria

- Can you manage change?

- What is your reaction to change?

- Among the skills necessary to manage change, which ones can you improve?

- Do you let things happen or make them happen?

Change and innovation

Change and Innovation

- Commutātio [-onis, f.]
- Immutātio [-onis, f.]
- Mutātio [-ōnis, m.]
- Emendātio [-ōnis, f.]
- Correctio [-onis, f.]
- **Innovatio** [-onis, f.] ✔
- Varietas [-atis, f.] – change of weather
- Vicissitudines [-um, f.] – change of luck

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Change and innovation

- Innovation triggers change (always)
- Change doesn’t always trigger innovation
  - only if there is a positively reaction
  - only if change becomes an opportunity

Innovation =

positive management of change

Change and innovation

- Managing change requires learning and adaptation
- **Ability (and time) to learn and adapt** are the main constrains to effective innovation

Innovation ***and growth***

Innovation and growth

Innovation is the central issue in economic prosperity

(Michael Porter)

Innovation is the central issue in economic prosperity

- ✓ Classical growth theory, Malthus
- ✓ Harrod-Domar growth model
- ✓ Neoclassical growth theory (Exogenous), Solow
- ✓ New growth theory (Endogenous), Romer, Lucas
- ✓ Mass Flourishing and Indigenous innovation, Phelps

*All growth theories find consensus on **innovation as a key generator of economic growth***

- ✓ *technological development*
- ✓ *technological progress*
- ✓ *knowledge/human capital*
- ✓ *indigenous/grassroots innovation*

1. Start ups vs. existing firms

2. Technological innovation vs. *low cost* innovation

3. Hi-tech innovation vs. innovation in traditional sectors

1. Startup vs. existing firms

- Number of startups is a sign of **dynamism** and source of **disruptive innovation**
- However, what matters is the number of **winners** (start-ups have a **high mortality rate**)
- Existing firms (incumbents) generate in aggregate more growth than startups

2. Technological innovation vs. *"low cost"* innovation

- Innovation based on R&D
- *"Low cost"* innovation
  - Process
  - Design
  - Use of new materials
  - Distribution and logistics
  - Pricing
  - Organization and management
  - Corporate governance

3. Hi-tech innovation vs. innovation in traditional sectors

- Sectors considered highly innovative: tech, hi-tech, biotech, biomed
- Potential of innovation in traditional sectors
  - credibility, tradition, education, training

✓ *Innovare senza passare per la Silicon Valley*

✓ ***Innovate without going through Silicon Valley***

(ASP Alumni conference, 2011)

Disruptive innovation

Process whereby a smaller company with fewer resources is able to successfully challenge established incumbent businesses

Clayton Christensen

- Typically
  - simpler products and services
  - initially at lower margin
  - targeting new and smaller markets

Sustaining vs. disruptive innovation

- **Disruptive** innovation
  - **Creates an entirely new market**
  - Easier to be embraced and initiated by **small firms** (values, cost structure, flexible and intuitive decision process, focus on markets too small for leaders)
    - Discount broker (Charles Schwab) vs. full-service
    - Introduction of the smartphone ✔
- **Sustaining** innovation
  - **Sustains customers by improving products and services in existing market**
  - Usually developed and introduced by **industry leaders**
    - Cash Mngmt Account (Merrill Lynch) allows customers to write checks against their equity accounts
    - Incremental innovation in cell phones and electronics ✔

Examples of disruptive innovation

zipcar

New market

- car rental by the hour

NETFLIX

New market

- (mailing and) online sales of content

Are these disruptive innvation?

- new markets ?

- low margin ?

Disruptive innovation

- Creates **new market** and **value network**
- Disrupts existing market and value network, displacing established firms and products

✓Steamboats	Sailing ships
✓High speed train	Short distance flights
✓Plastic	Metal, wood, glass
✓Smartphone	Telephone, PC, cameras
✓Digital photo, music, media	Chemical photos, traditional music, TV, CD, DVD
✓Wikipedia	Traditional encyclopedias
✓Ultrasound	Radiography

Disruptive innovation

- The ability to leverage **excess capacity** (even in an existing market) may be disruptive and may create new value

✓Uber	Taxi
✓Airbnb	Hotel
✓Whatsapp	Telcos
✓Waze	TomTom
✓MOOCs (1)	Colleges
✓Transferwise	Western Union

(1) MOOC: Massive Open Online Course

Revolutionary vs. disruptive innovation

- Not all revolutionary innovations are disruptive
  - Automobile (late ‘800)
    - revolutionary but not disruptive
    - too expensive
    - did not disrupt horse-drawn vehicles
  - Ford Model T (1908)
    - disruptive of the transportation market
    - low price
    - mass-produced

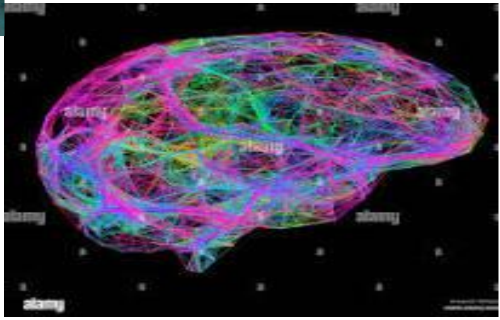
The new frontiers of innovation and international cooperation (or new fronts of conflict)?

Arctic

Space

Underwater

Artificial Intelligence



Summary Conclusions

Managing change is crucial

Suffering

Resistance

Resilience

Promoting

Managing change is primarily a matter of mindset

Change and innovation

- Innovation triggers change (always)
- Change doesn’t always trigger innovation (only if positively managed)
- **Innovation = positive management of change**
- Managing change requires learning and adaptation
- **Ability (and time) to learn and adapt** are the main constrains to effective innovation

Innovation and growth

- Innovation is the central issue in economic prosperity
- Measuring innovation is very important yet it is very difficult
- Number of start ups is a sign of dynamism but existing firms provide in aggregate more growth
- R&D drives innovation but there are many ways (also “low cost”) to innovate
- Hi-tech sectors are innovative but innovation in traditional sectors may provide significant growth

Disruptive innovation

- Creates **new markets** and **value networks**
- **Disrupts existing market and value network**, displacing established firms and products
- Ability to leverage **excess capacity** (even in existing market) may be disruptive and may create new value

Key innovation drivers

- Passion, curiosity
- Creativity
- Structure and discipline of thought
- Framework
- Competition (*fear factor*)
- Innovative dynamism
- Interaction
- Ideas (generation and circulation)

Thank you

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